TULSI RAJNIKANT PATEL

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EDUCATION

Arizona State University *Aug 2023 – May 2025*
Master of Science in Computer Science *Tempe, AZ | GPA: 4.00/4.00*

Nirma University *Jul 2019 – May 2023*
B.Tech in Computer Science & Engineering | Minor in Finance *Ahmedabad, India*

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Swift, SQL, JavaScript, TypeScript, Golang, HTML/CSS
ML/AI Frameworks: TensorFlow, PyTorch, Hugging Face, LangChain, Scikit-learn, OpenCV, Scikit-image
Cloud & DevOps: AWS (S3, DynamoDB), Docker, Kubernetes, Snowflake, Snowpark Container Services
Databases: MySQL, PostgreSQL, MongoDB, MongoDB Atlas, Snowflake Vector Store
Web Frameworks: React, Node.js, Django, Flask, Streamlit
Courses: Artificial Intelligence, Machine Learning, Natural Language Processing, Generative AI, Computer Vision, Cloud Computing, Data Structures & Algorithms, Database Management, Computer Networking, Software Engineering, Web Technologies

WORK EXPERIENCE

Machine Learning Engineer | Alkermes Inc., Waltham, MA (Remote) *Mar 2026 – Present*
- Developing EEG-based treatment prediction **machine learning models** (Elastic Net, Random Forest, LightGBM) on 5,000+ patient recordings, applying PCA-based dimensionality reduction to cut feature space by 70% while retaining 95% of variance, improving model training speed by 3x.
- Engineered end-to-end model evaluation infrastructure with cross-validation, SHAP-based explainability reports, and data visualization dashboards, collaborating with a team of 4 data scientists to ensure FDA-compliant, reproducible outputs via **MLflow**.

Machine Learning Engineer Intern | Alkermes Inc., Waltham, MA *Jul 2025 – Feb 2026*
- Spearheaded development of a clinical report automation app using **Snowflake AI Cortex**, integrating **Large Language Models** (GPT-5, Llama-3.1-70B, Mistral Large) to auto-generate 10+ structured report sections, cutting drafting time by 80% for a team of 8 medical writers.
- Architected a secure **RAG pipeline** using **Snowflake Vector Store** and **AWS S3**, embedding 1,000+ proprietary clinical documents to enable sub-second semantic retrieval with >90% summary accuracy, serving as a core MLOps component of the production system.
- Designed an LLM-powered **chat assistant** allowing medical writers to refine report content via natural language prompts, reducing revision cycles by 65% and receiving a 4.7/5 usability score across 3 departments.
- Containerized the AI application using **Docker** and deployed on Snowpark Container Services with auto-scaling, supporting 15 concurrent users.

Research & Development Intern | Sahajanand Technologies Pvt. Ltd., Surat, India *Jan 2023 – May 2023*
- Developed a **Computer Vision** pipeline for automated diamond inclusion detection using **OpenCV** and **Scikit-image**, processing 500+ gemstone images with contour-based and segmentation techniques, improving detection accuracy to 93% versus an 80% manual baseline.
- Applied **preprocessing** techniques (histogram equalization, noise reduction) to handle variable image quality, and designed **image-processing pipelines** to extract facet and table zones, reducing region segmentation errors and cutting per-stone inspection time from 8 to under 2 minutes.
- Integrated **AWS DynamoDB** as the metadata store for grading results, enabling structured querying across 1k+ graded diamonds and reducing data retrieval latency.

Undergraduate Research Intern | Nirma University, Ahmedabad, India *Jun 2022 – Dec 2022*
- Orchestrated a large-scale **NLP** data pipeline using SNScrape and Google Translate API to collect 600K+ multilingual tweets from 523 politicians across 12 languages for a university-funded digital humanities project.
- Fine-tuned **Large Language Models (BERT, RoBERTa)** and **deep learning models (CNN, LSTM, GRU)** for sentiment analysis and gendered language classification, with RoBERTa achieving 86% accuracy, a 14-point gain over the CNN baseline.
- Delivered interactive **Power BI** dashboards visualizing NLP-derived gendered language patterns, contributing to research that won 2nd Prize at Digital Humanities in Oxford Week, judged by Oxford University and Alan Turing Institute researchers.

Software Engineer Intern | Avioryn, Surat, India (Remote) *Jun 2022 – Aug 2022*
- Co-developed core backend modules for 'CleanZone', a laundry logistics platform, using **Node.js** and **MongoDB**, enabling real-time order tracking and reducing lost-order support queries across partner operations.
- Implemented OTP-based authentication via the **2Factor API,** QR code generation for product handoff, and a password recovery module in **JavaScript**, securing transactions and streamlining re-engagement for 30+ partner accounts.

PROJECTS

AI-Powered Clinical Assistant
- Built an end-to-end **RAG pipeline** using **OpenAI** text-embedding-3-small, semantically indexing 60K+ clinical records from the MedQA dataset and storing 1.2M+ embeddings in **MongoDB Atlas**, reducing query response time from 4s to 1.5s via HNSW indexing.
- Deployed Mistral-7B-Instruct via **vLLM** with a HIPAA-compliant prompt template, achieving 91.5% answer accuracy on a 500-question evaluation set and keeping hallucination rate below 5%.

Bon Appétit – Full-Stack Restaurant Platform
- Programmed a full-stack restaurant website using **React, Node.js** and **MongoDB** to handle menu management and reservations, designing an interactive **Bootstrap** UI that supported 100+ menu items and processed 200+ monthly bookings.
- Secured the admin panel with **JWT** authentication and configured automated email notifications via **Nodemailer API**, streamlining reservation management and reducing manual communication overhead.